Module: 1 – Linux Server – Understand and Use Essential Tools

1. What is the minimum number of partitions you need to install Linux?

Ans : Minimum **1 partition** is required (root / partition). Commonly, Linux also uses a **swap partition**.

2. Explain about chmod command

Ans : chmod stands for **change mode**. It is used to change file or directory permissions (read, write, execute).

3. How to check Linux memory utilization

Ans : Command:

```
free -h  or  top
```

4. Use grep to search for specific patterns in files

Ans : Example:

```
grep "hello" file.txt
```

This searches for the word **hello** in file.txt.

5. Get connecting on a Linux server by SSH

Ans : Command:

```
ssh username@ip-address
```

Example:

```
ssh root@192.168.1.10
```

6. Create 5 files in /tmp directory, then use tar and gzip to bundle and compress files

Ans : Commands:

```
cd /tmp
touch file1 file2 file3 file4 file5
tar -czvf files.tar.gz file1 file2 file3 file4 file5
```

7. Describe the root account

Ans : The **root account** is the administrator account in Linux. It has full permissions to access and modify all files, users, and system settings.

8. What is shell?

Ans : A **shell** is a command-line interface that allows users to interact with the Linux operating system.

9. What is Linux?

Ans : Linux is an **open-source operating system** based on Unix, used for servers, desktops, and embedded systems.

10. What is Bash?

Ans : Bash (**Bourne Again Shell**) is the default command-line shell in many Linux systems.

11. You have a new empty hard drive that you will use for Linux. What is the first step you use?

Ans : Create a **partition** on the hard drive using tools like fdisk or parted.

12. Write the Linux command to show the current working directory

Ans : pwd

13. Write the Linux command to get help with various options

Ans : man command_name

Example:

man ls

14. Write the Linux command to display what all users are currently doing

Ans : who

15. Write the Linux command to get information about the operating system

Ans : uname -a

or

cat /etc/os-release

16. Write the Linux command to create a hard link of a file

Ans : ln file1 file2

17. Write the Linux command to create a soft link of a file as well as directory

Ans : ln -s source destination

Example:

ln -s /home/test/file.txt linkfile

ln -s /home/test folderlink

18. Write the Linux command to search for a specific pattern in a file

Ans : grep "pattern" filename

19. Write the Linux command to show the use of basic regular expressions using grep command

Ans : Example:

grep "^hello" file.txt

This finds lines starting with **hello**.

Another example:

grep "[0-9]" file.txt

This searches for numbers in a file.